Project Proposal: Comprehensive Client Requirements & Web Application Solution

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1. Executive Summary

This document serves as the formal project proposal and comprehensive requirements specification for the "Abha Pharmacy Digital Transformation Project." Developed for a real-world client operating a local retail pharmacy, this document outlines the severe operational inefficiencies of their current manual record-keeping system, as well as the friction experienced by their daily customers.

To address these issues, we propose the development of a robust, dual-sided web application. This digital solution will completely modernize the business by providing the pharmacist with an automated inventory and sales management dashboard, while simultaneously providing the pharmacy's customers with a convenient online portal for prescription uploads, medicine availability checks, and home delivery requests.

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2. Detailed Client Profile & Business Background

2.1 Client Identification

- **Client Name:** Mr. Abhinav Kumar (Uncle)
- **Business Name:** Abha Pharmacy
- **Location:** Urban Retail Sector
- **Business Nature:** Retail Pharmacy, Prescription Medicines, and General Healthcare Supplements.

2.2 Business Environment and Scale

Abha Pharmacy has been a cornerstone of the local community for over five years. The business operates seven days a week, processing an average of 150 to 200 individual customer transactions daily. The pharmacy carries an extensive inventory of over 2,500 distinct SKUs, ranging from fast-moving consumer goods to highly regulated prescription medications.

Currently, the store is managed by Mr. Abhinav and two assistant pharmacists. Despite the high volume of foot traffic and the critical nature of the products being sold, the entire operational backbone of the pharmacy relies on archaic, manual data management techniques. Furthermore, their relationship with their customers is strictly limited to in-person, over-the-counter interactions.

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3. Analysis of the Current System (The Problem Space)

Upon interviewing the client and observing the daily workflow, our team identified several critical vulnerabilities that are hindering business growth, causing financial loss, and negatively impacting customer satisfaction.

3.1 Internal Problems (Pharmacist's Perspective)

1. **Inventory Tracking and the Expiration Crisis:** Medicines are perishable goods. In the current spreadsheet system, tracking the expiration date of every single strip of medicine is nearly impossible. The pharmacist often discovers expired medicines only during physical audits. This leads to massive financial losses when expired medicines must be discarded.
2. **The Stock-Out Dilemma:** Because sales are recorded on paper, the inventory sheet is not updated in real-time. The pharmacist frequently runs out of critical, fast-moving medicines because the system does not warn them when stock is running low.
3. **Manual Accounting Overhead:** At the end of the day, Mr. Abhinav spends over two hours manually tallying the physical sales ledger with a calculator to determine the day's revenue, a process highly prone to human error.

3.2 External Problems (Customer's Perspective)

1. **Wasted Trips and Frustration:** Currently, a sick patient must physically travel to the pharmacy just to find out if a specific prescribed medicine is in stock. If the pharmacy is experiencing a stock-out (as mentioned above), the customer's time is entirely wasted, leading to frustration and loss of loyalty.
2. **Long Waiting Queues:** Customers dropping off complex prescriptions often have to stand in long queues while the pharmacist manually searches the back room to gather all the required medications.
3. **No Remote Refill Options:** Elderly patients or those with chronic conditions who require monthly refills of the exact same medication must physically visit the store every single month, as the pharmacy has no digital infrastructure to process remote refill requests.

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4. Comprehensive Client Requirements

Based on our extensive problem analysis, we have mapped out exactly what the client requires from a new digital system to solve both internal operational issues and external customer friction.

4.1 Business Objectives

1. **Modernize the Customer Experience:** Provide a digital avenue for customers to interact with the pharmacy remotely.

2. **Reduce Wastage:** Decrease financial loss from expired medicines by proactively identifying them.
3. **Optimize Ordering:** Ensure zero stock-outs of critical medicines through automated tracking.
4. **Streamline Operations:** Eliminate manual, end-of-day mathematical tallying.

4.2 User Requirements (User Stories)

For the Pharmacist (Admin):

- *As a pharmacist*, I want a dashboard that automatically shows me which medicines are expiring next month, so I can return them to the supplier in time.
- *As a business owner*, I want the system to alert me when a medicine's stock drops below 10 units, so I can reorder it before it runs out.
- *As a pharmacist*, I want a digital checkout screen to quickly process walk-in sales and automatically generate a digital receipt.

For the Customer (End-User):

- *As a customer*, I want to search the pharmacy's inventory online from my phone, so I know if they have my medicine before I leave my house.
- *As a customer*, I want to take a photo of my doctor's prescription and upload it to the website, so the pharmacist can prepare my order before I arrive.
- *As an elderly patient*, I want to request a home delivery for my monthly blood pressure medication directly through the web portal.

4.3 Functional Requirements (System Features)

1. **Dual Authentication System:** Secure login portals distinguishing between Admin (Pharmacists) and Users (Customers).
2. **Admin Inventory Module:** Interface to register new medicines, update stock quantities, and record expiration dates.
3. **Admin POS & Alert Module:** Digital checkout interface that automatically deducts stock upon sale and triggers visual alerts for low stock/expirations.
4. **Customer Search Catalog:** A public-facing search bar allowing customers to check the real-time availability and price of OTC medicines.
5. **Prescription Upload Engine:** A secure form allowing registered customers to upload image files (JPG/PNG) of their prescriptions and attach delivery/pickup instructions.
6. **Order Fulfillment Dashboard:** An admin screen where the pharmacist can view incoming online prescription uploads, approve them, and notify the customer when the order is ready for pickup.

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5. Proposed Solution: The Dual-Portal Web Application

To fulfill every requirement outlined above, our group will build a comprehensive Web Application. By utilizing modern web technologies (HTML, CSS, JavaScript for the interface, and Python/Flask for the backend logic), we will completely eliminate the archaic manual processes currently failing the client.

Here is exactly how the Web Application solves the identified problems:

5.1 Solving Internal Problems (The Admin Portal)

- **Eliminating Expirations and Stock-Outs:** The days of manual ledgers are over. When the pharmacist uses the digital POS module to ring up a customer, the Python backend instantly deducts that item from the central inventory. If the stock drops below a predefined threshold, or if the backend detects an expiration date approaching within 30 days, the web app triggers a bright red "Action Required" notification on the pharmacist's main dashboard. This totally automates inventory management and prevents financial loss.
- **Automated Accounting:** Because every transaction is recorded digitally, Mr. Abhinav will no longer need a calculator. A dedicated "Reports" page will instantly calculate daily, weekly, and monthly revenue with perfect mathematical accuracy.

5.2 Solving Customer Problems (The Customer Portal)

- **Ending Wasted Trips:** The web application will feature a customer-facing portal. If a customer is looking for a specific cough syrup, they can simply type it into the website's search bar. The web app will read the exact same real-time inventory data the pharmacist uses, instantly telling the customer "In Stock" or "Out of Stock." This transparency builds immense customer trust.
- **The "Click and Collect" Prescription Feature:** To solve the issue of long queues and waiting times, the web app will include a secure "Upload Prescription" button. A sick patient can snap a photo of their doctor's note from bed and submit it. The pharmacist receives a ping on their Admin dashboard, prepares the medicine bag during quiet store hours, and clicks "Ready for Pickup." The customer simply walks in, grabs their bag, and leaves, drastically improving the customer experience.
- **Remote Refills:** Registered users can view their past purchase history on the web app and click a single "Request Refill" button for their chronic medications, optionally requesting home delivery. This transitions Abha Pharmacy from a simple local shop into a modern, service-oriented healthcare provider.

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6. Conclusion

The current operational model at Abha Pharmacy—relying exclusively on manual ledgers and fragmented physical processes—is fundamentally unsustainable for a growing business. It actively causes financial leakage through expired stock, frustrates customers with frequent stock-outs, and consumes hours of administrative time daily. Furthermore, the lack of any digital interface completely alienates the modern, tech-savvy consumer who expects remote convenience, especially in healthcare.

By developing this custom, dual-sided Web Application, our group will provide Mr. Abhinav Kumar with a comprehensive digital transformation. The pharmacist gains total, automated control over their inventory and sales analytics, eliminating human error and drastically reducing waste. Simultaneously, the customers gain a seamless, friction-free portal to manage their health needs remotely, from checking stock to uploading complex prescriptions.

This project bridges the gap between traditional community healthcare and modern digital convenience, resulting in a more profitable business for the client and a vastly superior experience for the patients.